

2026 Comprehensive Road Maintenance Operation Procedures

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- **Purpose:**

The purpose of this Standard Operating Procedure (SOP) is to establish consistent, safe, and efficient operational practices for all maintenance activities conducted by the Marin County Department of Public Works – Roads Division. This document will provide a unified framework for planning, executing, documenting, and all maintenance tasks to ensure public safety, regulatory compliance, asset preservation, and high-quality service delivery.

- **Scope:**

- Pavement maintenance and repair
- Drainage systems and stormwater infrastructure
- Vegetation management and tree operations
- Traffic control and work zone management
- Signage and roadway striping maintenance
- Heavy equipment uses and maintenance
- Shoulder work and roadbed stabilization
- Storm event preparation and emergency response
- Slide/slip-out mitigation
- Customer service requests and field inspections
- Documentation, reporting, and communication

- **Responsibilities:**

Roads Superintendent

- Oversees SOP implementation and compliance
- Manages budgets, scheduling, and work priorities
- Approves and processes invoices and time sheets
- Works with Fleet on vehicle acquisition
- Coordinates with other divisions on projects/needs
- Conducts meetings with road & sign shop supervisors
- Reviews employee evaluations and provides support
- Communicates professionalism and teamwork

Road & Sign Shop Supervisors

- Direct and monitor work crews
- Enforce safety protocols and approve traffic control setups
- Ensure proper use of equipment and materials
- Reviews daily equipment logs and work reports
- Provides support to staff and document events as needed
- Manages inventory and works with accounting on best practices
- Works directly with Superintendent on current events

Maintenance Staff

- Perform maintenance tasks according to procedure
- Maintain safe work practices and proper PPE use
- Completes daily equipment reports
- Report hazards, equipment issues, and work completion
- Communicates with constituents and supervisors daily

Contractors

- Comply with all County standards and field direction
- Maintain required licensing, insurance, and safety compliance

- **Equipment and Materials**

- Dump trucks, transport, backhoes, loaders, graders, rollers, dozer, paving machine, mowers, pickup, stencil truck, striping truck. Auger truck, street sweepers
- Asphalt delivery and compaction tools
- Chainsaws, chippers, and vegetation equipment
- Traffic control devices (cones, signs, barricades, flagger gear, CMS Boards)
- Hand tools (shovels, rakes, compaction tampers, brooms chainsaws, pole saws, blowers)
- Storm response gear (pumps, sandbags, debris clearing tools,

- **Materials used may include:**

- Hot asphalt, pothole mix, concrete, aggregate base
- Culvert pipe and drainage hardware
- Signs, posts, delineators, striping materials
- Erosion control materials
- Fuel, lubricants, and consumables

- **Safety Requirements:**

All personnel shall comply with:

- Cal/OSHA regulations
- County safety policies
- MUTCD-compliant traffic control standards
- Equipment safety manuals
- PPE requirements including high-visibility vests, hard hats, gloves, boots, eye/hearing protection

- **Procedures:**

1. Pavement Maintenance

- Conduct Road inspections to identify cracks, trip hazards, potholes, or subbase failure
- Perform pothole repairs using hot mix or cold patch
- Apply crack sealant where required
- Conduct skin patching, edge repairs, and dig-outs as needed.
- Ensure proper compaction and site cleanup

2. Drainage Maintenance

- Inspect culverts, ditches, and inlets
- Remove debris and flush where necessary
- Repair or replace damaged pipes, headwalls, or grates
- Restore flow and stabilize eroded areas
- Remove any obstructions in curb and gutter leaves, sticks etc.

3. Vegetation & Tree Work

- Brush roadside shoulders and clear sight lines
- Trim overgrown vegetation obstructing signs streetlights, stoplights or drainage
- Remove hazardous trees using proper tree-felling procedures and equipment
- Chip and dispose of vegetation at local dump sites.

4. Traffic Control

- Determine appropriate control per MUTCD
- Set up signs, cones, barricades, flaggers
- Inspect work zone for safety before starting work
- Monitor and adjust flow as needed

5. Signs & Striping Maintenance

- Replace damaged or missing signs
- Install new signs per County standards
- Maintain striping, markers, legends and crosswalks
- Clean signage obstructed by vegetation or debris

6. Shoulder & Roadbed Stabilization

- Grade shoulders to restore proper drainage
- Add aggregate or soil where settlement has occurred
- Compact and shape shoulders for safety and runoff control

7. Storm Response and Emergency Operations

- Patrol roads during storm events and report road closers
- Clear debris, fallen trees, rockslides
- Sandbagging, pumping, and flooding mitigation
- Assist emergency services as needed
- Document conditions and report hazards immediately

8. Slip-Out Response

- Assess danger zones from a safe distance
- Install temporary traffic control
- Remove loose material when safe
- Install sandbags or k-rail as needed
- Coordinate with engineering for permanent repair design when needed

9. Customer Requests & Inspections

- Respond to service requests in a timely manner
- Document findings, actions, and follow-up needs
- Communicate with superintendent or engineering as needed

Documentation and Reporting

All work activities must be documented, including:

- Daily crew logs, truck cards
- Work orders and completed tasks
- Material and equipment usage
- Incident or near-miss reports
- Storm activity logs
- Photographic documentation when applicable